

Theory of Mechanism and Machine

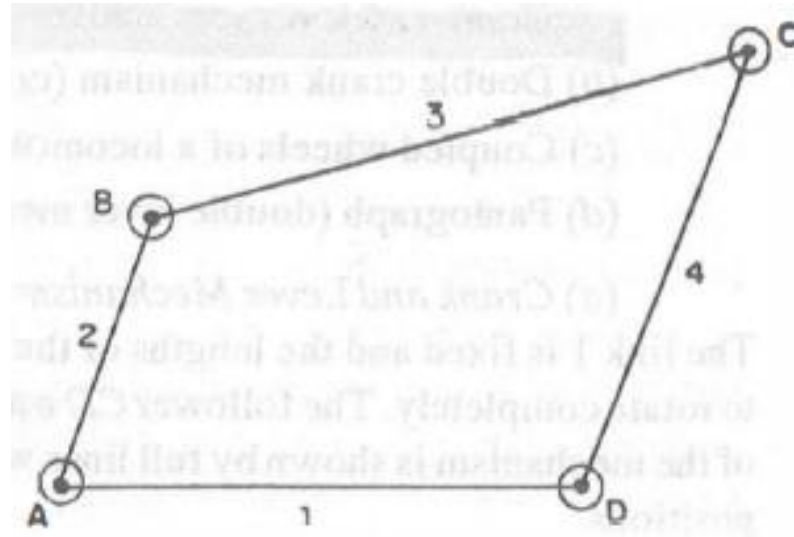
(Week-2)

INVERSION OF MECHANISMS:

- A kinematic chain becomes a mechanism when one of its links is fixed. This method of obtaining different mechanisms by fixing different links of a kinematic chain is called inversion of the mechanism.
- The relative motion between the various links is not altered as a result of inversion, but their absolute motion with respect to the fixed link may alter drastically.
- *Different Types of Kinematic Chains and Their Inversions:*
- The simplest kinematic chain is a chain consisting of four kinematic pairs, each pair being a sliding pair or a turning pair. The important types of kinematic chains with four kinematic pairs, are
 - (i) *Four bar chain*,
 - (ii) Single slider crank chain, and
 - (ii,) Double slider crank chain.

Inversions of a Four-Bar Chain Mechanism:

- This is the simplest kinematic chain. It consists of four rigid links which are connected in the form of a quadrilateral by four pin joints as shown in Fig.



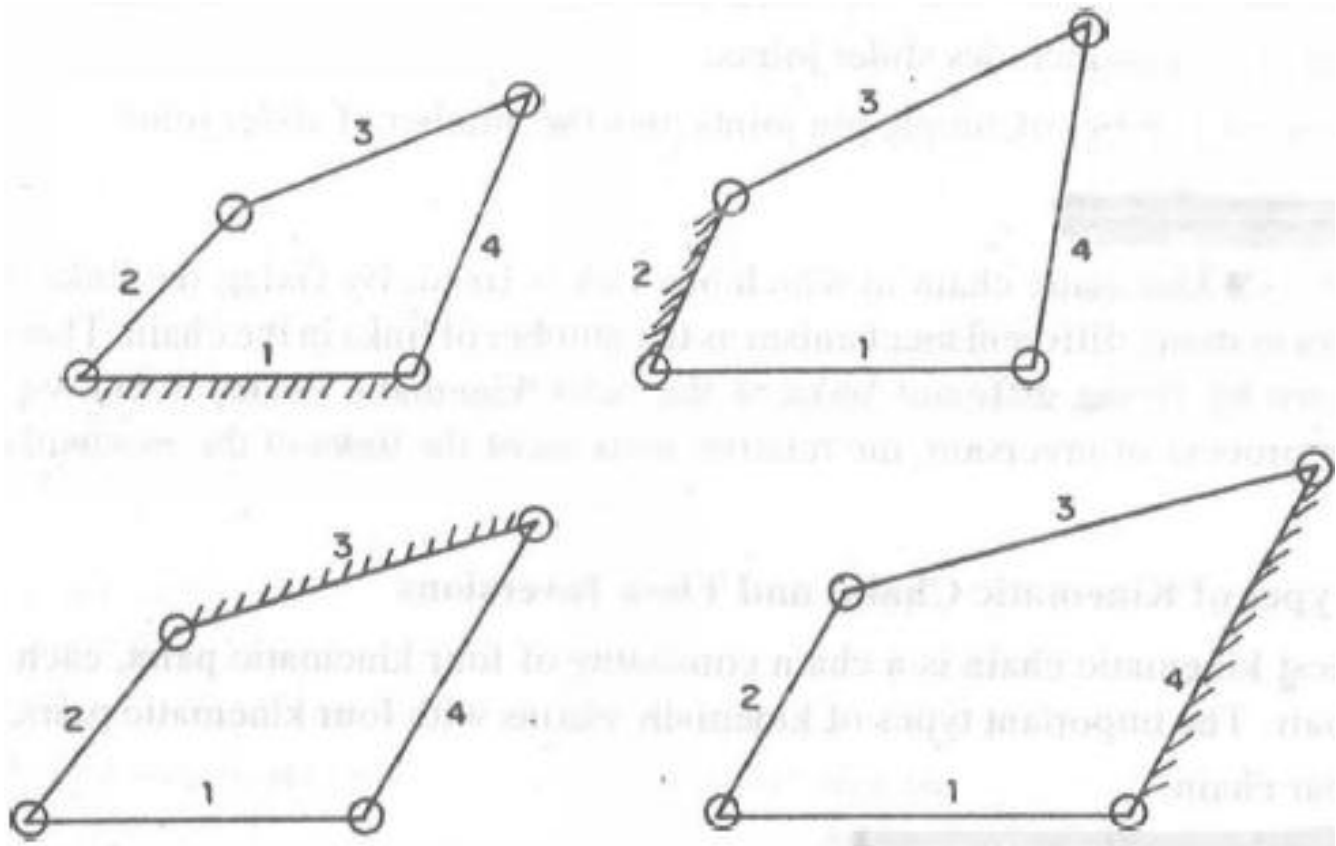
- Link 1 and link 2 forms first turning pair, link 2 and link 3 form second turning pair, link 3 and link 4 forms third turning pair and link 4 and link 1 forms fourth turning pair.

A link that makes complete revolution is known as crank.

The fixed link is known as frame of the mechanism.

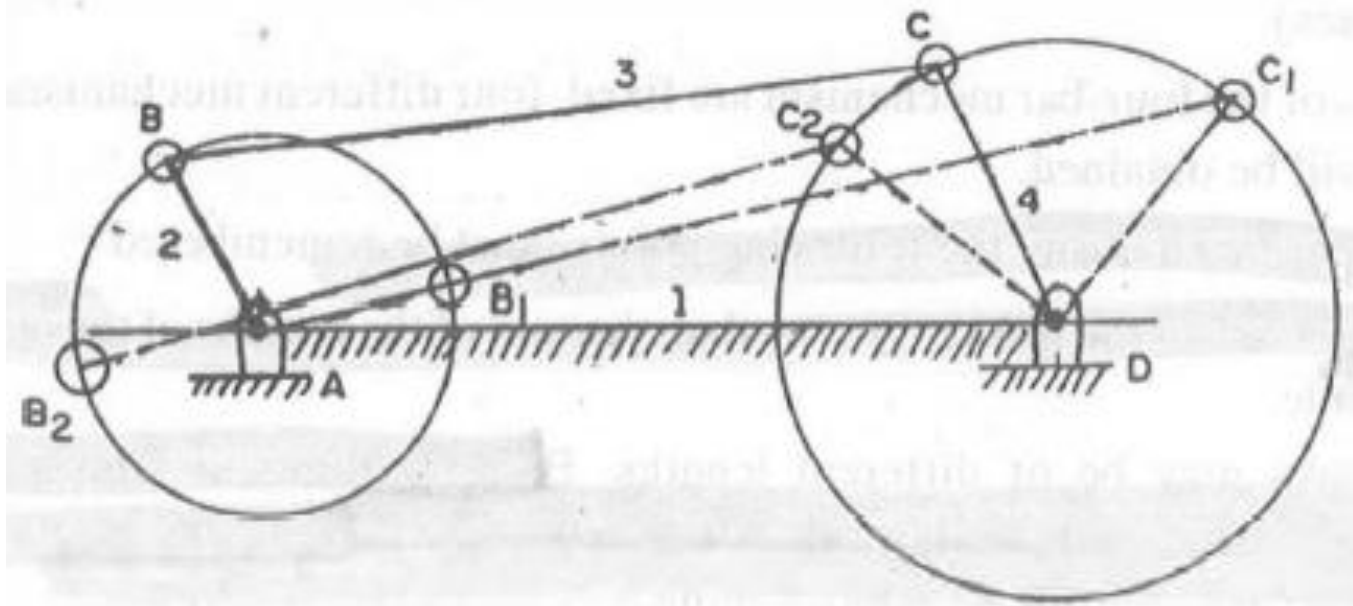
The link opposite to the fixed link is known as coupler or connecting rod.

The fourth link is known as lever or rocker (if it oscillates) or an another crank (if it rotates).

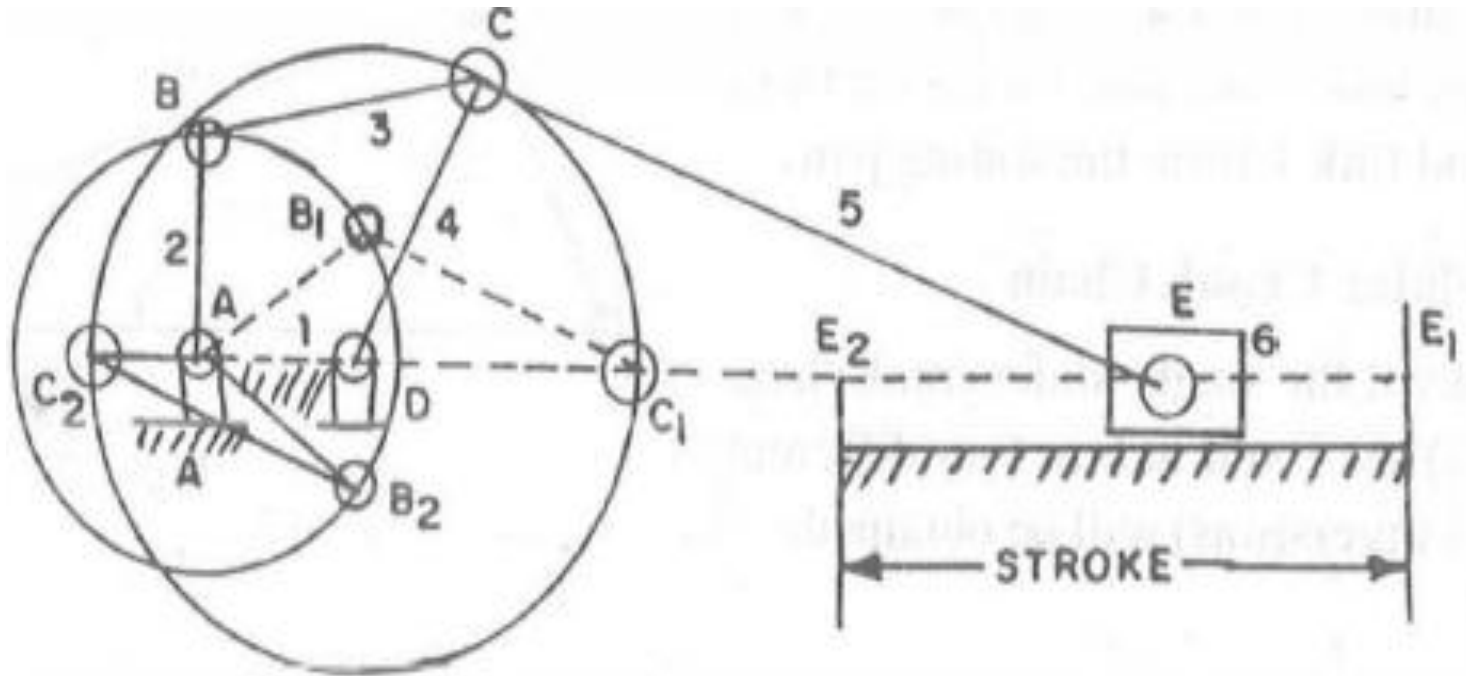


- Applications of Four Bar Chain.
- The followings are the important applications of four bar chain
- (a) Crank and lever mechanism (oscillatory motion)
- (b) Double crank mechanism (complete rotation of the crank and the follower)
- (c) Coupled wheels of a locomotive (double crank)
- (d) Pantograph (double lever mechanism)

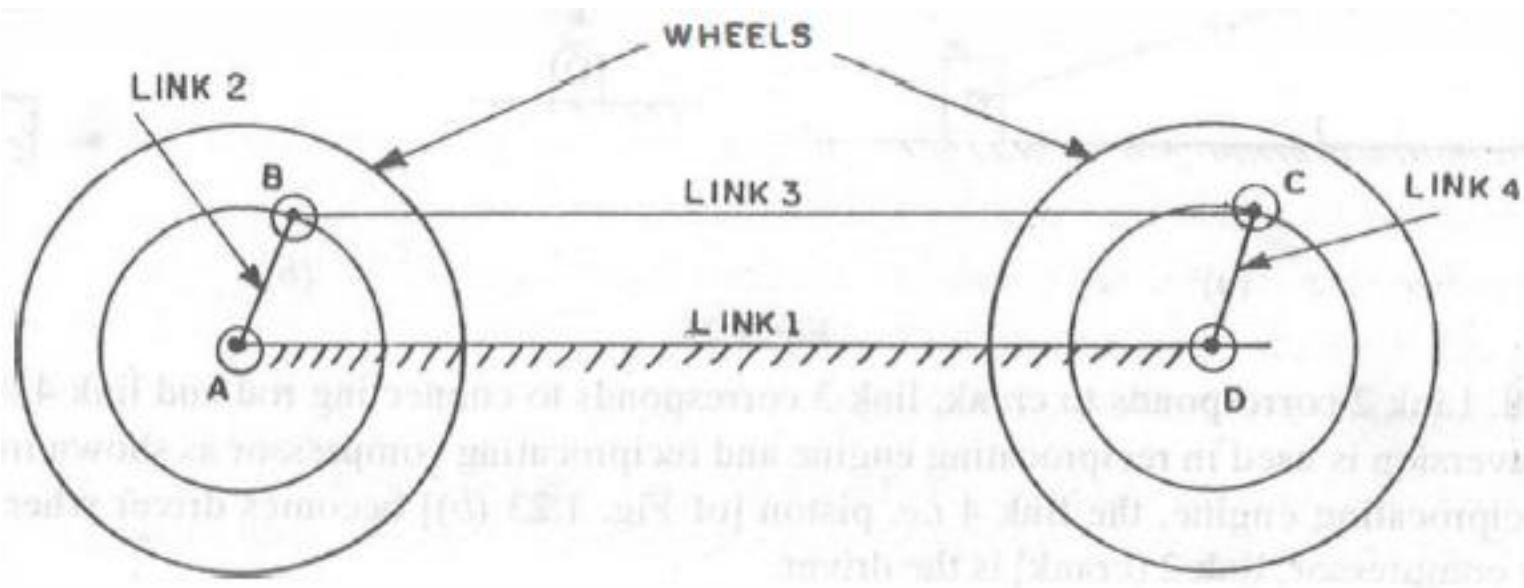
- **Crank and Lever Mechanism (oscillatory motion).**
- The link-1 is fixed and the lengths of the links-2,3 and 4 are proportionate in such a way that crank BA is able to rotate completely.
- The follower CD only oscillate from C1 to C2 as shown in Fig.
- The initial position of the mechanism is shown by full lines whereas the dotted lines show the mechanism for two extreme different positions.



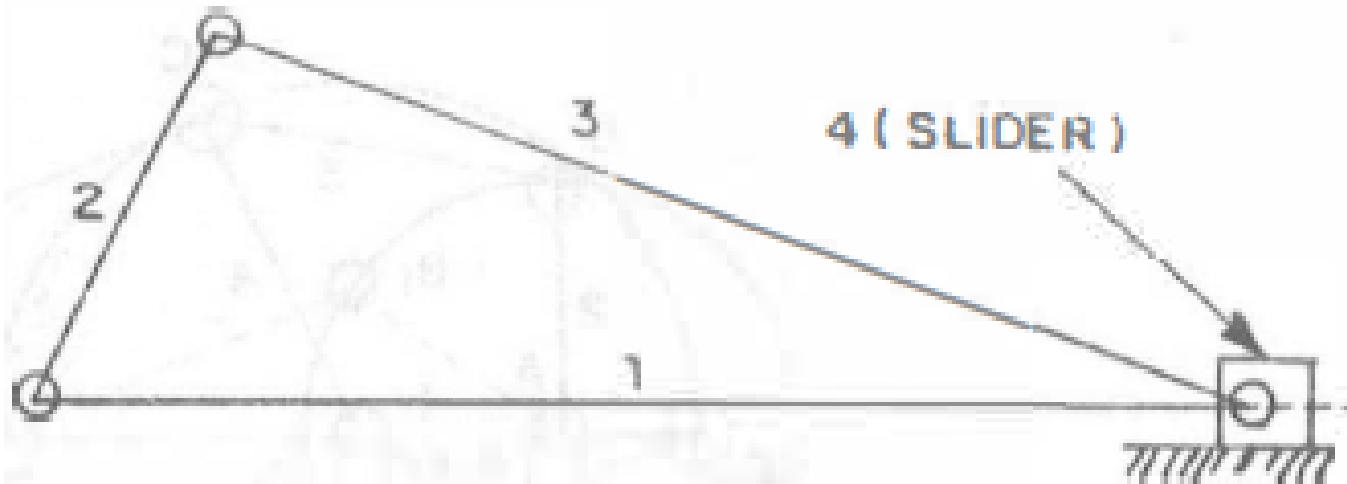
- *Double Crank Mechanism (complete rotation of the crank and the follower):*
- *The shortest link 1 is always stationary link. The sum of the shortest and the longest links should be less than the sum of the other two links.*
- *For a drag link quick return motion, link 5 and link 6 are also connected as shown in Fig. The initial position of the mechanism is shown by full lines i.e. ABCD along with CE. The two extreme positions (i.e. AB₁C₁D along with C₁E₁ and position AB₂C₂D along with C₂E₂) are shown by dotted lines.*



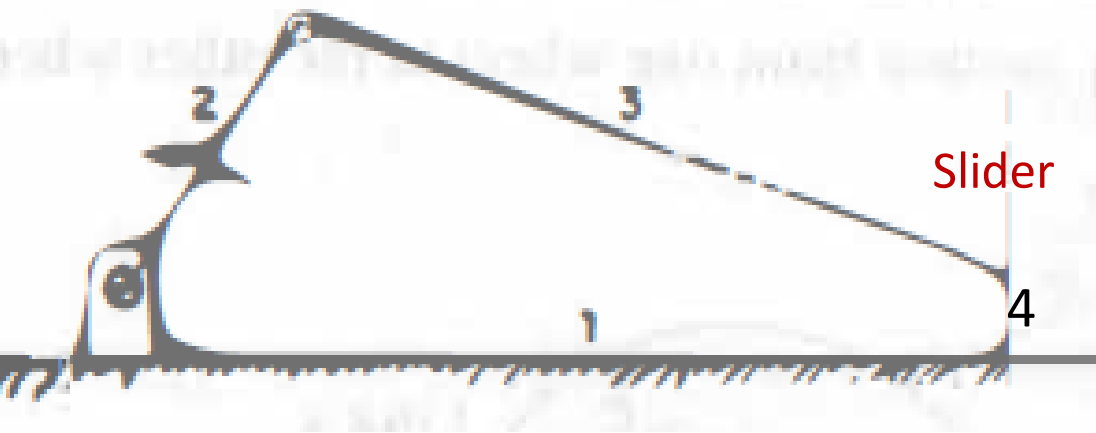
- *Coupled wheels of a locomotive (Double crank):*
- The below Fig. shows the coupled wheels of a locomotive which is also known as double crank mechanism. The links AB and CD are of equal lengths and act as cranks.
- These cranks are connected to the respective wheels. The link AD is a fixed link and link BC acts as a coupling rod. This mechanism is used for transmitting rotary motion from one wheel to the other wheel.



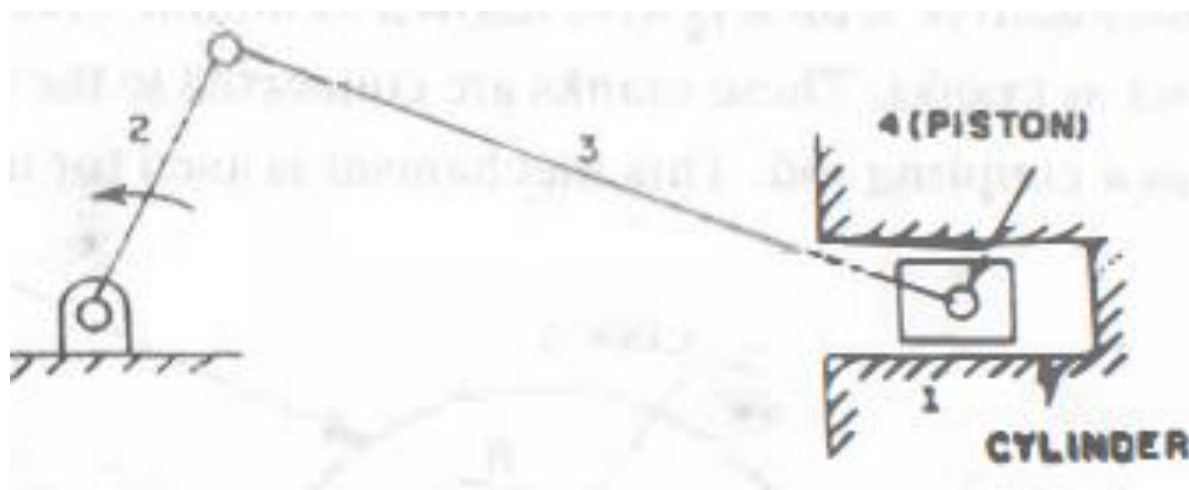
- **Single-Slider Crank Mechanism:**
- *It is the modification of basic four bar chain. It consists of four kinematic pairs out of which one is sliding pair and three arc turning pairs as shown in Fig.*
- *Link 1 and link 2 form one turning pair, link 2 and link 3 form the second turning pair, link 3 and link 4 form the third turning pair. Link 4 and link 1 form the sliding pair.*



- *Inversion of Single Slider Crank Chain:*
- If different links of the single-slider crank chain are fixed in turn four different mechanism (known as inversions) will be obtained.
- *First Inversion:*
- When link 1 is fixed, link 2 is made crank and link 4 is made slider, then first inversion of single slider crank chain is obtained. This inversion is shown in Fig.



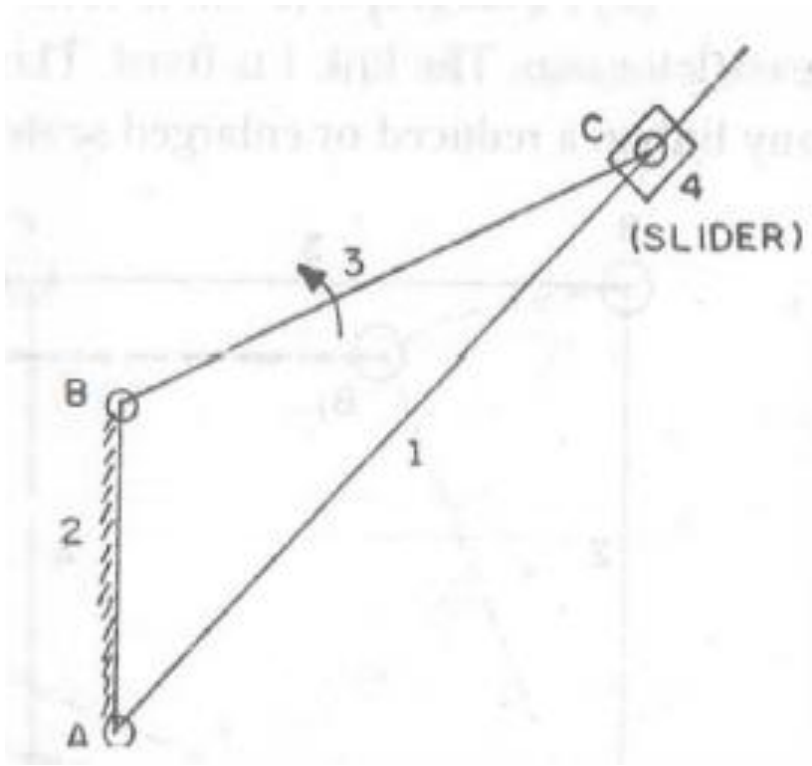
- Here the link 1 corresponds to the frame which is fixed. Link 2 corresponds to crank, link 3 corresponds to connecting rod and link 4 to corresponds to slider. This inversion is used in reciprocating engine and reciprocating compressor as shown in fig.



- In case of reciprocating engine, the link 4 i.e. piston becomes driver whereas in case of reciprocating compressor, link 2 (crank) is the driver.

Second Inversion:

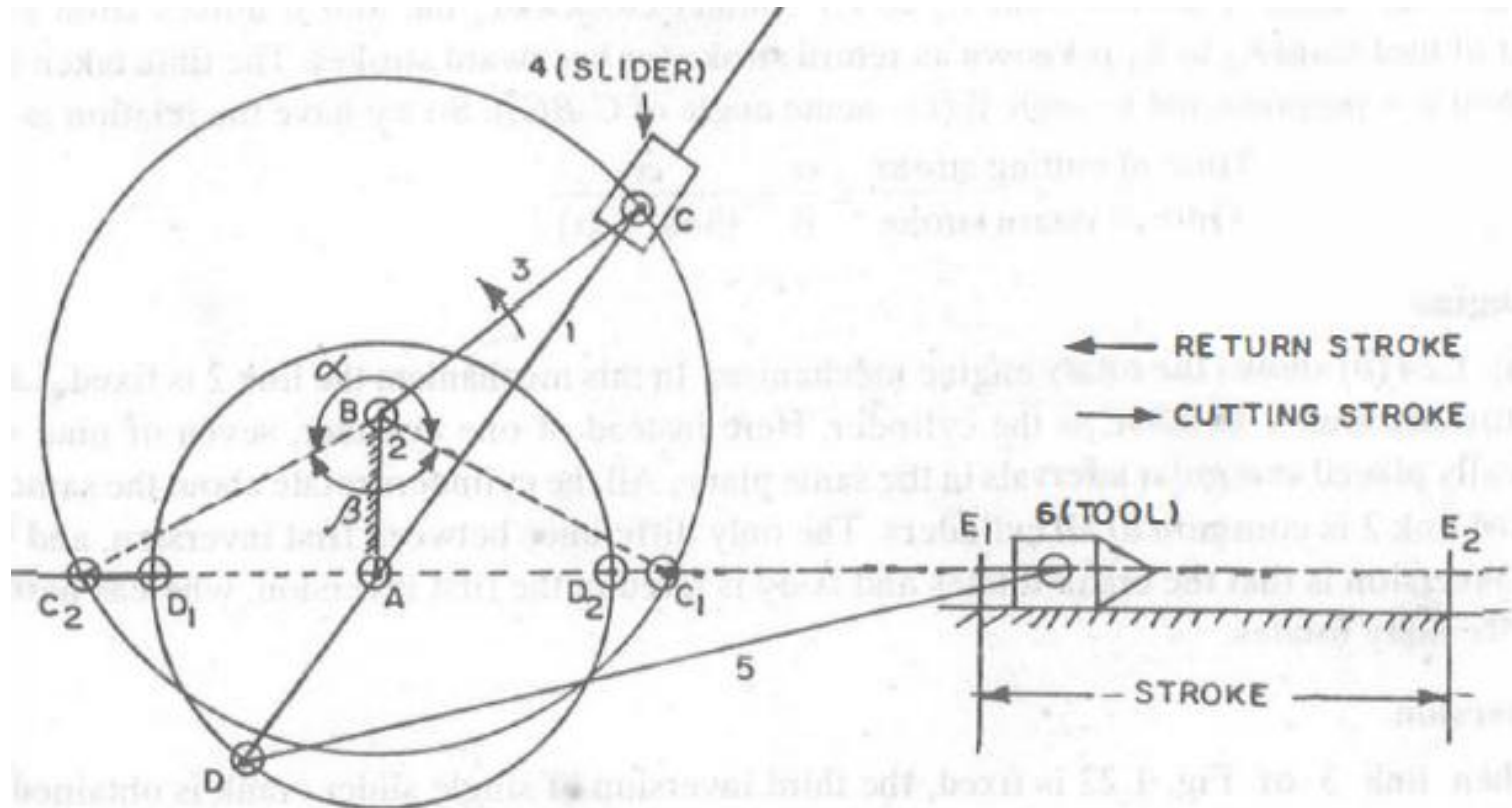
- When link 2 (or crank) of is fixed, the second inversion of single slider crank chain is obtained as shown in Fig.



Link-3 along with the slider at its end C, becomes a crank. Hence link-3 along with slider (link-4) rotates about B.

By doing so, the link-1 rotates about A along with the slider (link 4) which reciprocates on link-1. This inversion is used in whitworth quick-return mechanism and rotary engine.

- *Whitworth Quick-Return Mechanism:*
- This mechanism is used in workshops to cut metals. The forward stroke cuts the metal whereas the return stroke is idle. The forward stroke takes a little longer period whereas the return stroke takes a shorter period. The below Fig. shows this mechanism in which link 2 is fixed.

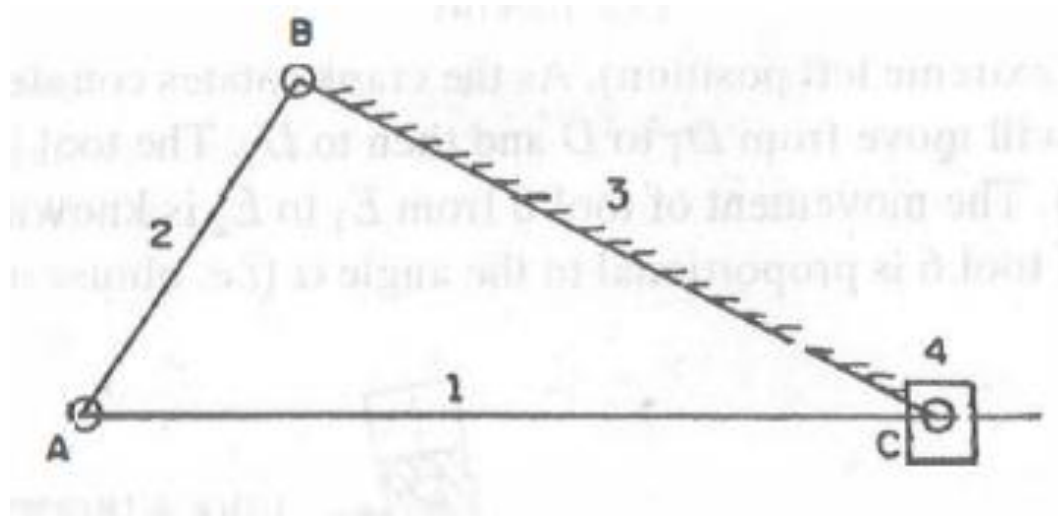


- The link-3 along with its slider (i.e. link 4) rotates in a circle about B. By doing so, the link-1 rotates about A along with the slider which reciprocates on link-1. On the link-1, produced downward there is a point D, where link 5 is connected.
- The other end of the link 5 is connected to the tool (link 6). The forward stroke of the tool cuts the metal whereas the return stroke is idle. The point D rotates in a circle about point A.
- The axis of motion of the tool passes through A and is perpendicular to AB, the fixed link. The crank-3 rotates in a counterclockwise direction.
- Let, initially the slider 4 is at C1 • The point D will be then at D1 and the tool- 6 will be at E1 (i.e. the extreme left position).
- As the crank rotates counterclockwise from point C1 to C and then to C2, the point D will move from D1 to D and then to D2.

- The tool (link 6) will move from E1 to E2 (i.e. extreme right position). The movement of tool 6 from E1 to E2 is known as forward stroke.
- The time taken for the forward stroke of tool-6 is proportional to the angle α (i.e. obtuse angle C1BC2).
- When the slider 4 moves from C2 to C1 counterclockwise, the tool 6 moves from E2 to E1. The movement of tool from E2 to E1 is known as return stroke (or backward stroke).
- The time taken for the return stroke of tool-6 is proportional to angle β (i.e. acute angle of C2BC1). So we have the relation as:

$$\frac{\text{Time of cutting stroke}}{\text{Time of return stroke}} = \frac{\alpha}{\beta} = \frac{\alpha}{(360^\circ - \alpha)}$$

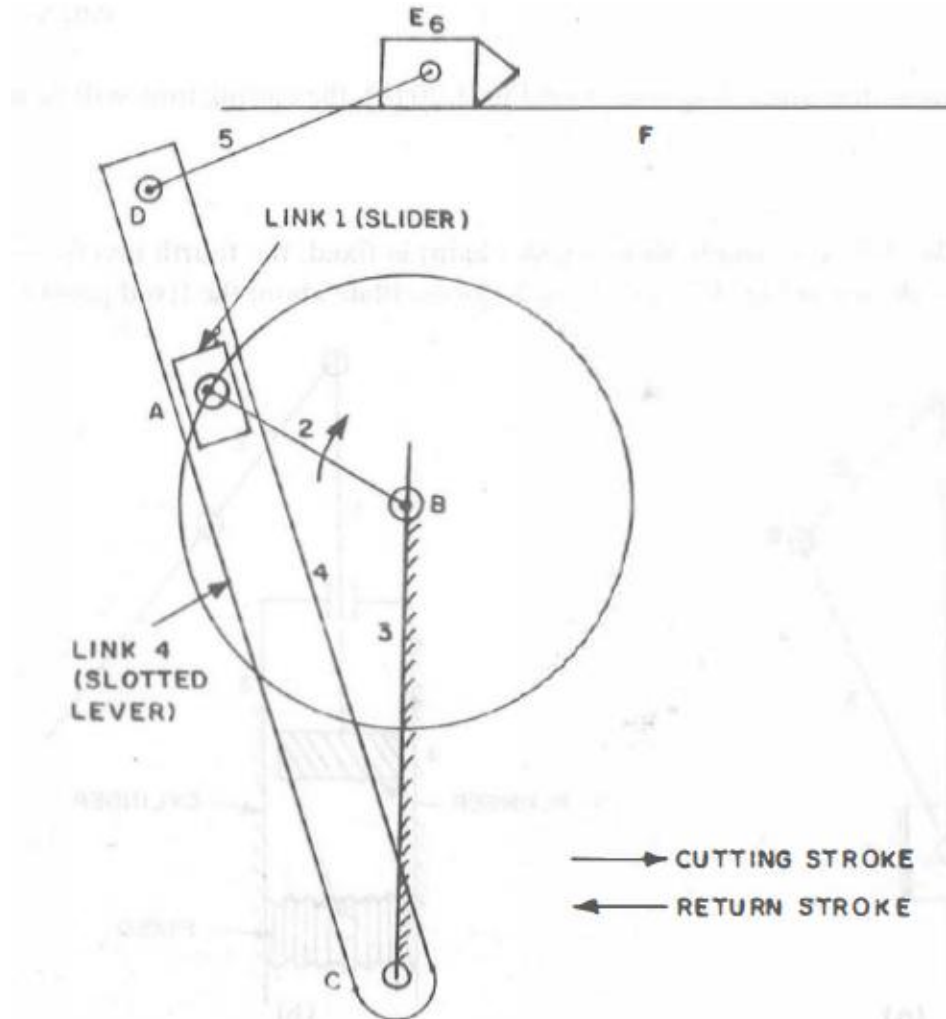
- *Third Inversion:*
- *When link 3 of is fixed as shown in below fig, the third inversion of single slider crank is obtained.*



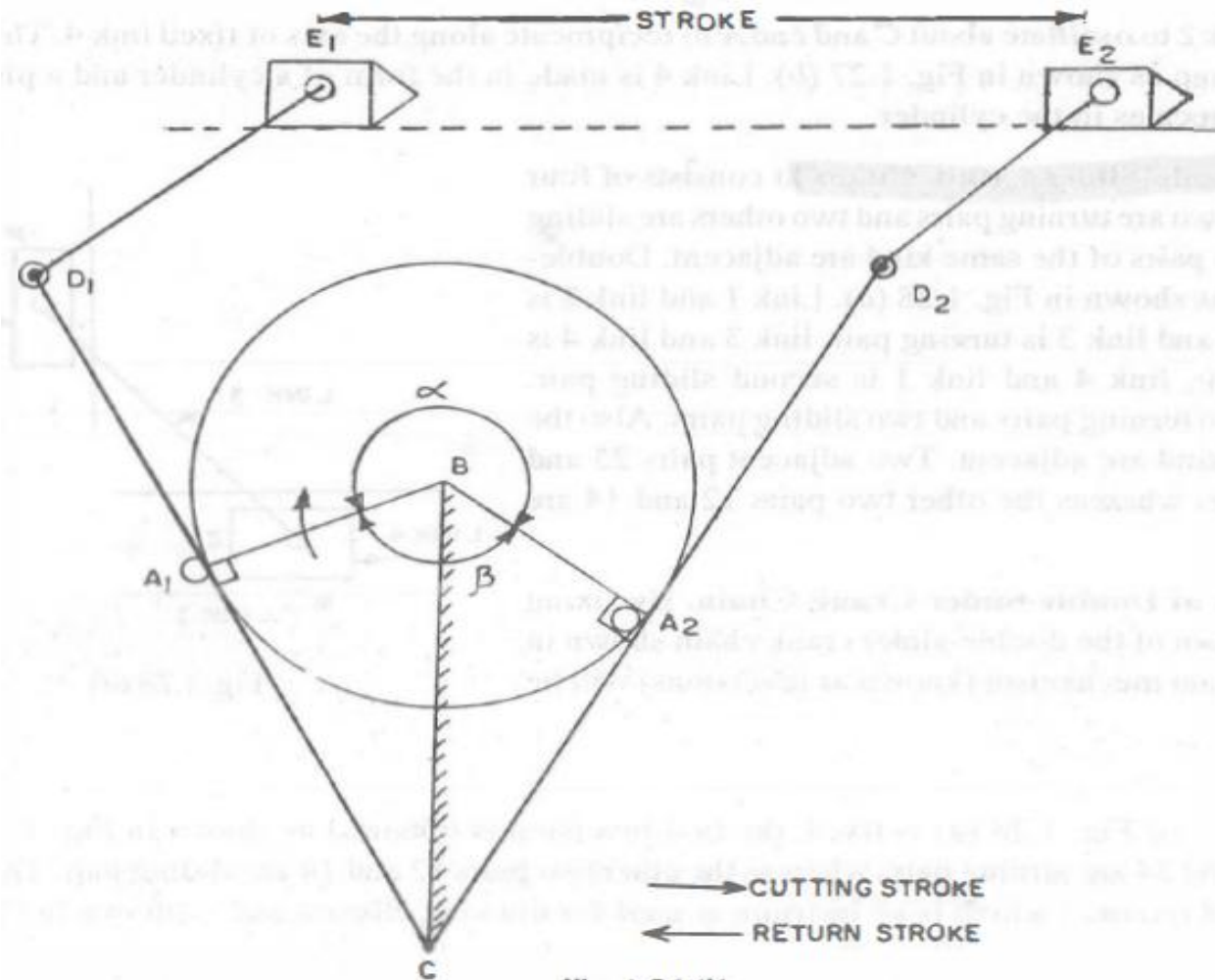
- *Link-2 acts as a crank and is rotating about point B. Link 4 oscillates.*
- *This inversion is used in oscillating cylinder engine and crank and slotted-lever mechanism.*

Crank and Slotted-Lever Mechanism:

- The below Fig. shows a crank and slotted lever mechanism, in which link-1 is a slider which slides in a slotted lever which is link 4. Link-3 is fixed and link-2 is a crank which rotates in clock-wise direction about point B in a circle. Due to this the link-4 oscillates about the point C.



- Link-4 is extended to point D to which link DE (i.e. link 5) is connected. At the end of link 5, cutting tool i.e. link-6 is attached. Link-6 is constrained to slide.
- The below Fig. shows the two extreme position of the crank.

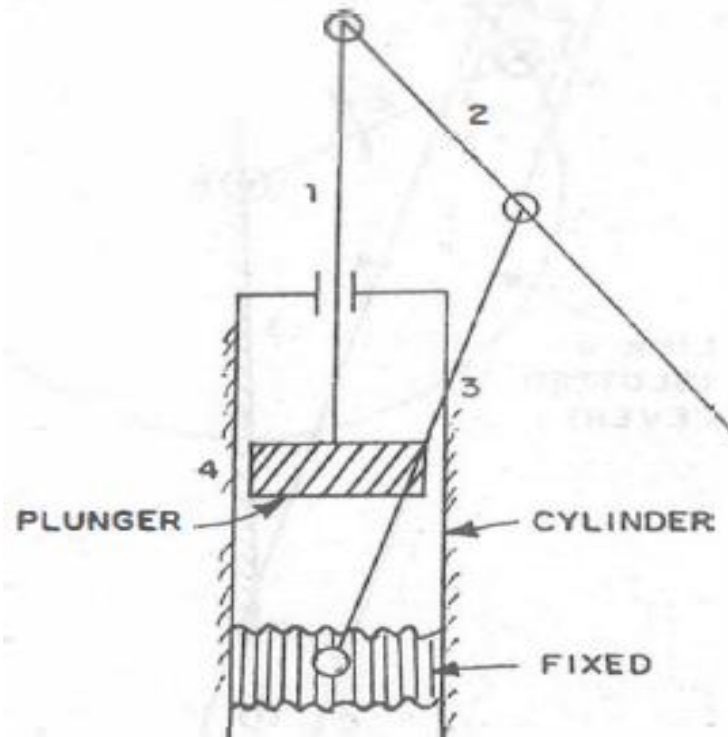
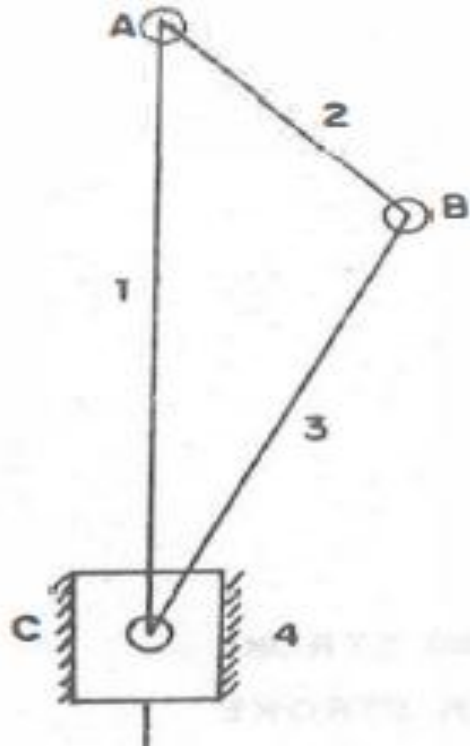


- The first position is when the crank (i.e. link 2) is at right angles to link 4 (or link 4 is tangential to crank radius circle at point A1). The remaining corresponding points in this position will be D1 and E1.
- The stroke of the cutting tool starts from point E1 . The crank is rotating in clock-wise direction. The end of the cutting stroke is marked by point E2, when again crank after having rotated through an angle α in the clock-wise direction is again at right angles to the link 4 at position.
- Hence the cutting stroke occurs when crank rotates through an angle α and return stroke occurs when crank rotates through an angle β or $(360 - \alpha)$ in the clock-wise direction. Since the crank has a uniform angular speed, therefore

$$\frac{\text{Time of cutting stroke}}{\text{Time of return stroke}} = \frac{\alpha}{\beta} = \frac{\alpha}{(360 - \alpha)}$$

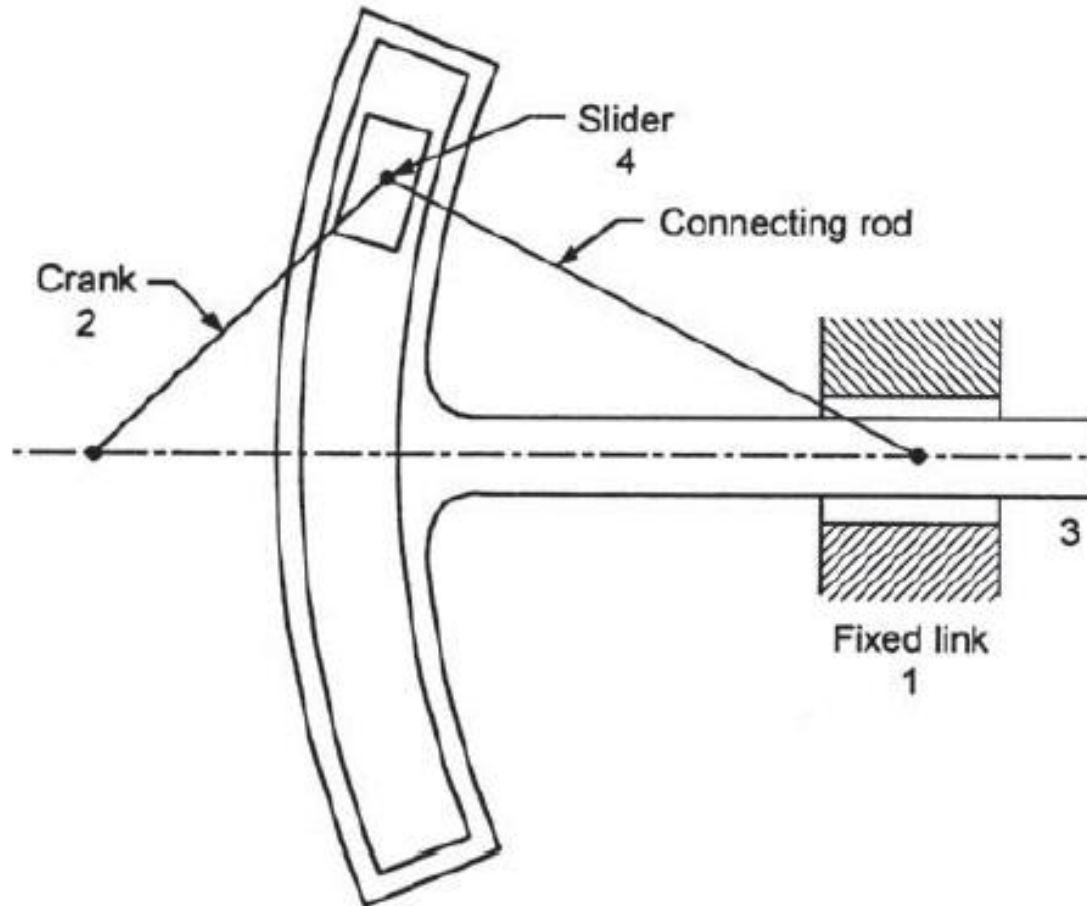
Fourth Inversion:

- When link-4 of i.e. single slider crank chain is fixed, the fourth inversion of single slider crank chain is obtained as shown in left Fig. Link-3 can oscillate about the fixed point C on link-4.
- This makes end B of link-2 to oscillate about C and end A to reciprocate along the axis of fixed link-4. This inversion is used in hand-pump as shown in right Fig. Link-4 is made in the form of a cylinder and a plunger fixed to the link-1, reciprocates in the cylinder.

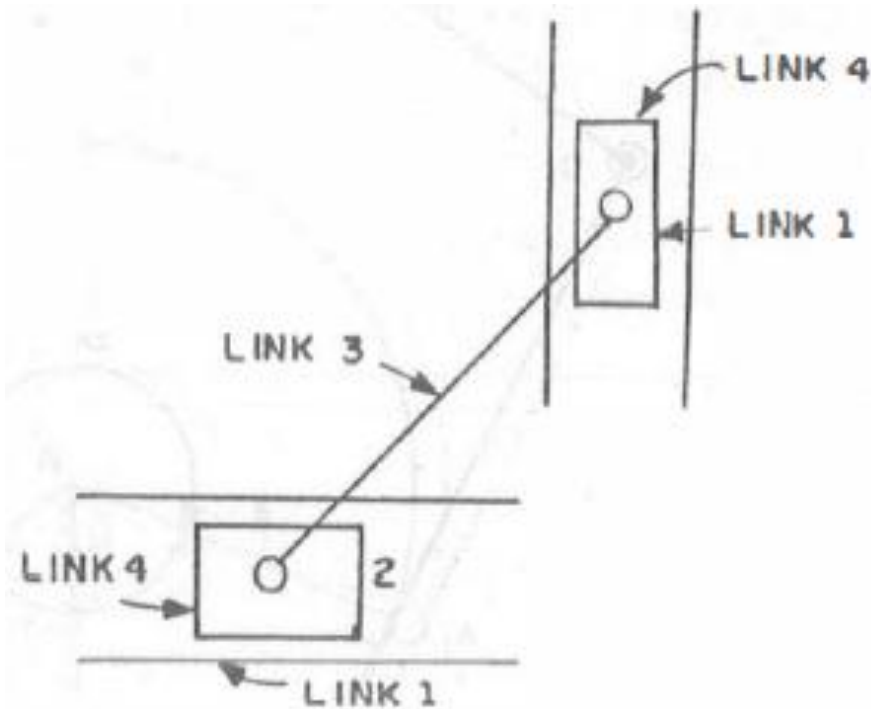


DOUBLE SLIDER-CRANK CHAIN:

- A kinematic chain consisting of two turning pairs and two sliding pairs is called double slider-crank chain as shown in Fig.



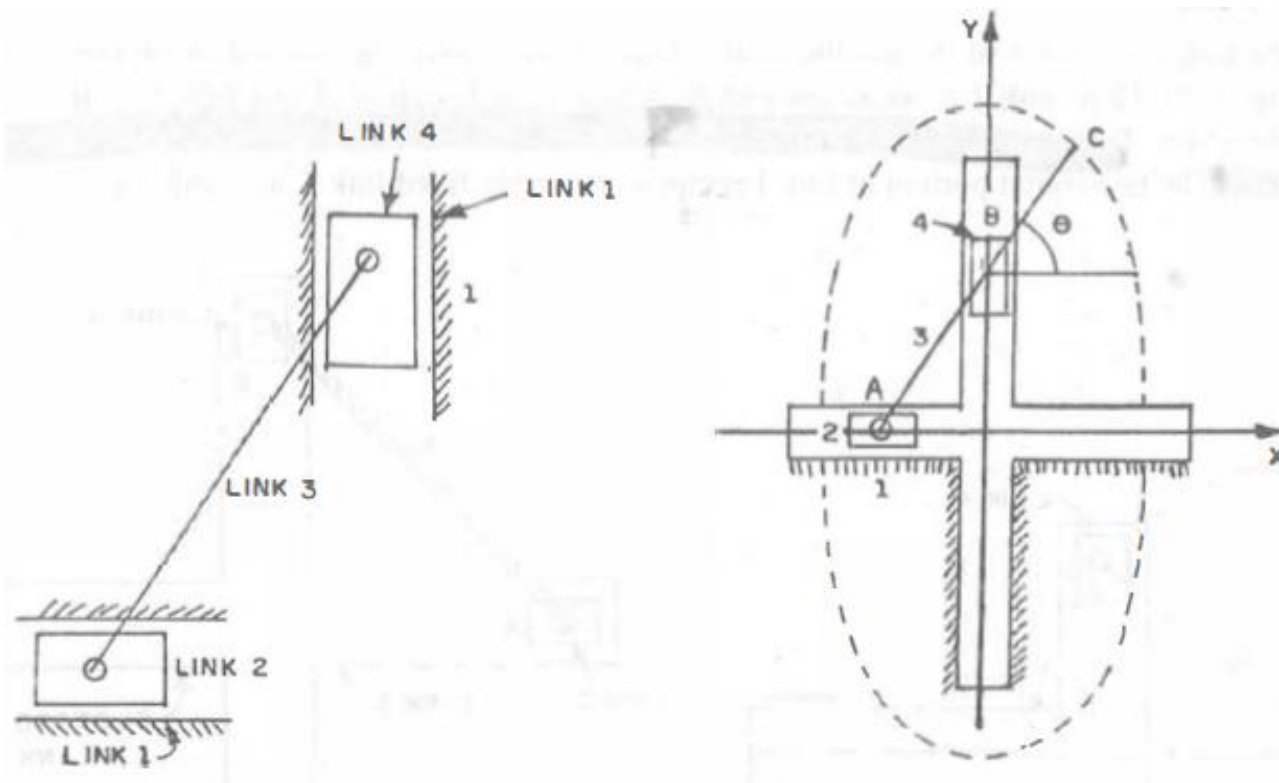
- *Double slider crank chain is shown in below Fig.*
- *Link -1 and link-2 is sliding pair, link-2 and link-3 is turning pair. Link -3 and link-4 is second turning pair, link-4 and link-1 is second sliding pair.*
- *Hence there are two turning pairs and two sliding pairs. Also the pairs of the same kind are adjacent. Two adjacent pairs 23 and 34 are turning pairs whereas the other two pairs 12 and 14 are sliding pairs.*



Inversions of Double-Slider Crank Chain:

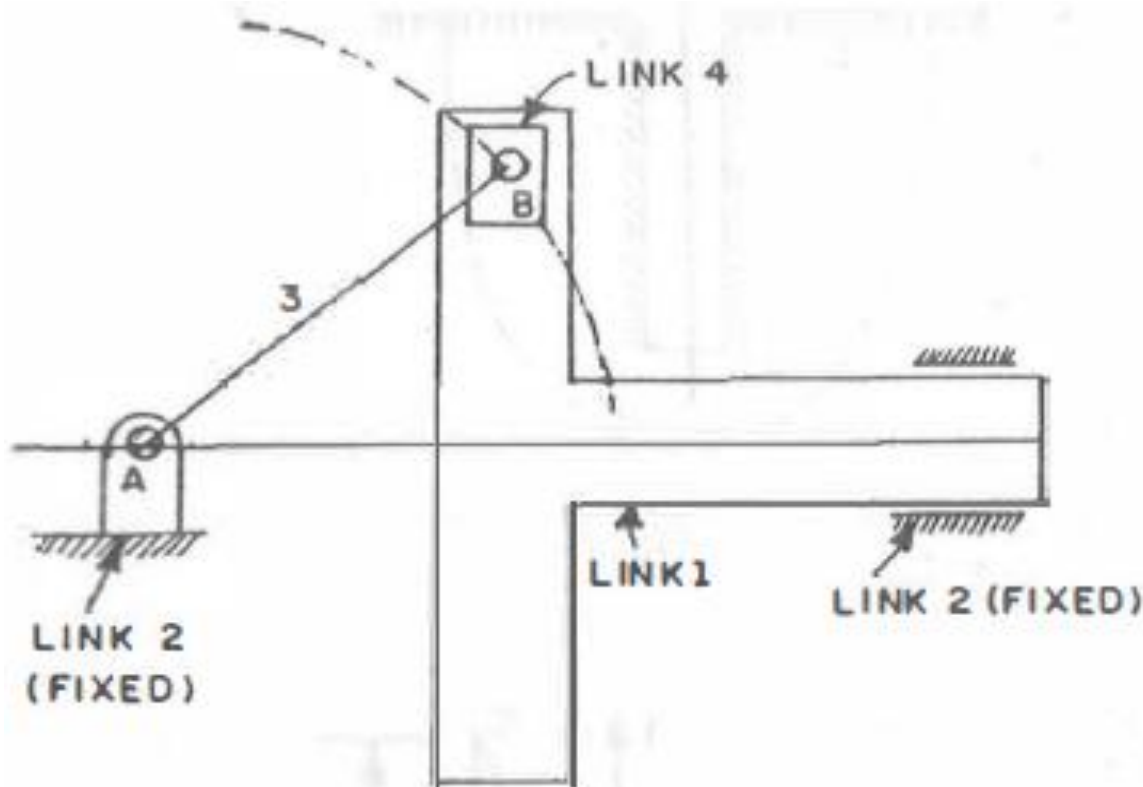
First Inversion:

When link I is fixed, the first inversion is obtained as shown in left Fig . Two adjacent pairs 23 and 34 are turning pairs whereas the other two pairs 12 and 14 are sliding pair. This inversion is used in Elliptical trammel which is an instrument used for drawing ellipses and is shown in Fig.



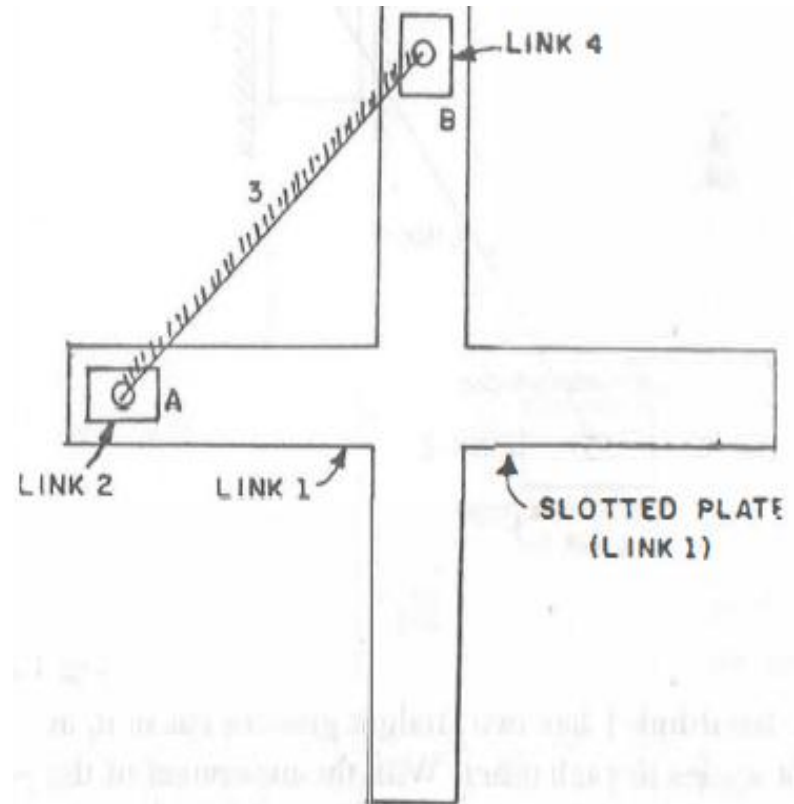
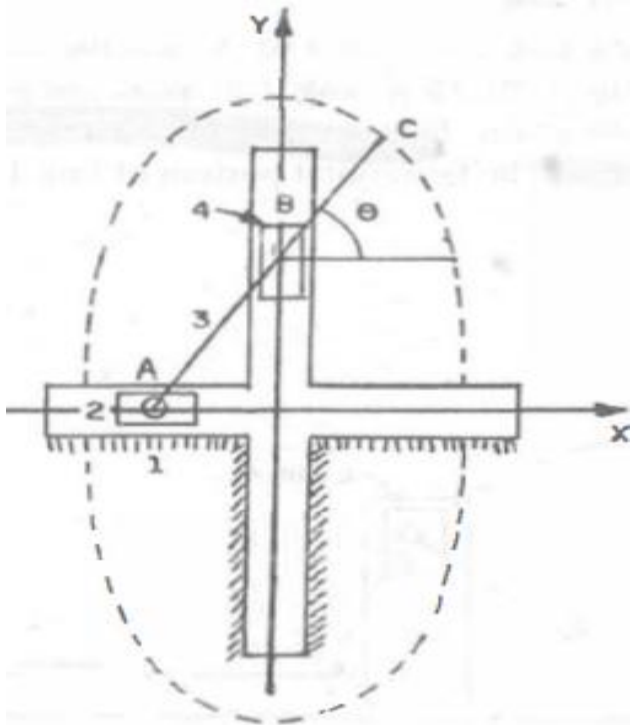
Second Inversion:

- When link-2 or link-4 of the double-slider crank chain is fixed, the second inversion is obtained as shown in Fig. Here link-2 is fixed and end B · Link-3 rotates about A and link-1 reciprocates in the horizontal direction. This inversion is used in Scotch (yoke, which is used to convert the rotary motion into a sliding motion. The horizontal portion of link I reciprocates in the fixed link-2 as crank 3 rotates.



Third Inversion:

- When link 3, of the double slider crank chain shown in left Fig is fixed and link-1 is free to move, third inversion is obtained which is shown in right Fig.
- In this case each of slide blocks (i.e. link-2 and link-4) can turn about the pins A and B. If one slide block (say link 2) is turned through a definite angle, the frame (i.e. link-1) and the other block (i.e. link 4) must turn through the same angle.



- This inversion is used in Oldham's Coupling as shown in Fig. which is used for connecting two parallel shaft when the distance between the two shafts is small. The two shafts to be connected have flanges at their ends which are rigidly fastened by forging to the shafts. These flanges forms links 2 and 4. Each of these links forms a turning pair with link 3.